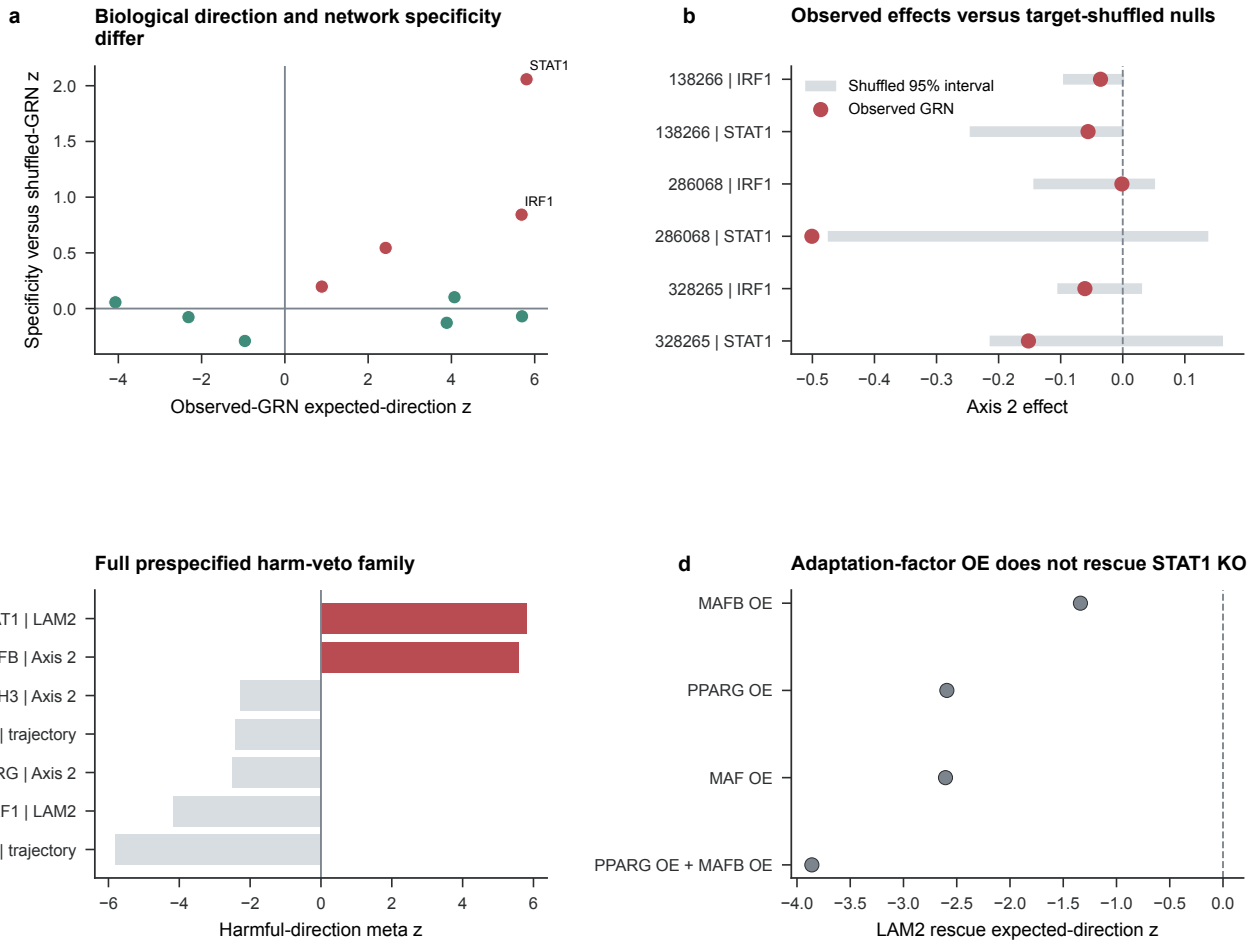
a
Biological direction and network specificity differ
STAT1
IRF1
Specificity versus shuffled-GRN z
Observed-GRN expected-direction z
b
Observed effects versus target-shuffled nulls
Shuffled 95% interval
Observed GRN
138266 | IRF1
138266 | STAT1
286068 | IRF1
286068 | STAT1
328265 | IRF1
328265 | STAT1
Axis 2 effect
c
Full prespecified harm-veto family
STAT1 | LAM2
MAFB | Axis 2
NR1H3 | Axis 2
IRF1 | trajectory
PPARG | Axis 2
IRF1 | LAM2
STAT1 | trajectory
Harmful-direction meta z
d
Adaptation-factor OE does not rescue STAT1 KO
MAFB OE
PPARG OE
MAF OE
PPARG OE + MAFB OE
LAM2 rescue expected-direction z